NLP AND BIG DATA · FINAL PROJECT

Torque Diagnostics

A technician chatbot for ABB ACS880 industrial drives. It answers a fault or a hardware question only from the indexed English manuals, cites the page, and withholds the procedure when the draft is not supported.

Ingest is batch | Answer is interactive | Low confidence refuses

S3 + Parquet
System of record

BM25 + vectors
How a fault code is found

NLI gate
Whether steps are shown

Concept

The equipment is an ABB ACS880 industrial drive. It controls the speed and torque of an electric motor on a pump, fan, conveyor, compressor, or hoist. It is not a generator, a computer, or the crane itself. A crane or winch appears only when that optional control program is loaded into the same drive.

The technician types a fault code, describes a symptom, or uploads a photo of the panel. The bot retrieves the matching manual pages, writes steps only from those pages, and labels the answer High, Medium, or Low. High shows the steps and the page. Low shows the passages and no procedure.

The primary control program uses one fault list for every ACS880 frame that runs it. Hardware questions are frame-specific. A question about an ACS880-17LC fan or Safe torque off circuit is answered from that frame's hardware manual. If that PDF is not in the lake, the answer is Low.

Eighteen drive types

The primary firmware manual names 18 type designations. Two power ranges of the same number are one device. AINLX and YINLX are firmware packages, not extra machines.

Type	What it is	Type	What it is
ACS880-01	Wall-mounted single drive	ACS880-31	Ultra-low harmonic drive
ACS880-04	Single drive module	ACS880-34	Ultra-low harmonic module
ACS880-04XT	Drive module package	ACS880-37	Ultra-low harmonic cabinet
ACS880-07	Cabinet single drive	ACS880-37LC	Liquid-cooled cabinet
ACS880-07LC	Liquid-cooled cabinet	ACS880-104	Inverter module
ACS880-07CLC	Liquid-cooled cabinet variant	ACS880-104LC	Liquid-cooled inverter module
ACS880-11	Regenerative wall-mounted	ACS880-107	Inverter unit
ACS880-14	Regenerative module	ACS880-107LC	Liquid-cooled inverter unit
ACS880-17	Regenerative cabinet	ACS880-17LC	Liquid-cooled regenerative cabinet

What the technician can ask

A code on the panel. The firmware fault-tracing chapter covers codes such as 2310 overcurrent, 2330 earth leakage, 2340 short circuit, 3210 DC link overvoltage, 5080 fan, and 7121 motor stall. For 2310 the published

checks include acceleration groups 23, 26, and 28, scaling parameters 46.01–46.03, the motor and motor cable, and parameter group 99 against the motor rating plate.

A symptom without a code. “Trips when I accelerate” can still retrieve the 2310 page. That answer is more often Medium, because the panel code was not in the question.

Which drive this is. The hardware manual explains the type designation. A label ending in -5 is the 380–500 V variant. Parameter 99.10, motor nominal power, has to match the motor plate. Start-up, maintenance, resistor braking, and Safe torque off are in scope when that section is in the indexed hardware PDF.

Out of scope. A code that is not in the indexed manual, a Siemens drive, a gearbox, or a photo whose code was misread. If OCR turns 2310 into 2340, the draft can match the short-circuit page. The technician still confirms the code on the panel.

Confidence gate

Three signals, one label. A contradiction on the entailment check overrides a strong retrieval score.

Label	Rule	What the technician sees
High	Strong hybrid hit, the draft is entailed, and every question entity is in the evidence	Steps, page citation, highlighted rows
Medium	Partial entity match, or entailment is neutral	Steps with caveats and the extra passages
Low	Weak retrieval, or the entailment model contradicts the draft	Passages only. No procedure.

Architecture

One worker builds the index. The chat request only reads it. The chat service never parses a PDF.

Ingest path

A manual is written once to S3, then parsed into page-level Parquet. Polars on the worker writes the code map, including the tracing page on each fault row, and the hybrid index. An upload of a manual is indexed. A photo of the panel is a question, not a new manual.

Upload PDF or photo → **FastAPI** → **S3 raw zone** → **SQS** → **Docker worker** → **Parquet lake**

then **Hybrid index**

Answer path

A typed fault or a photo of the panel is turned into entities, retrieved from that drive family’s index, and scored before any step is shown. The tracing page comes back on the chunk row.

Text or photo → **NER on the question** → **BM25 + embeddings**

then **Answer from passages** → **NLI + coverage** → **Cite or refuse**

On the request path

FastAPI, named-entity recognition, hybrid search, generation from the retrieved rows, and the entailment score. The target is a short answer while the machine is down. Retrieval is a lookup in the code map, the BM25 index, and the embedding index. It does not scan the PDFs.

On the worker

OCR, Polars, the Parquet write, and the hybrid index. OCR of a long scan stays on the worker. A photo of the panel is read first. After the code exists, retrieval is the same lookup.

What the request is allowed to do

The chatbot reads the code map, the BM25 index, and the embedding index that the worker already built. DuckDB reads the same Parquet for coverage checks, such as which fault codes have no indexed procedure. The request does not rebuild the lake.

Tech stack

The worker builds the lake and the index. The chatbot only reads them.

Layer	Tools
Chat and upload	FastAPI. A manual upload is indexed. A panel photo is a question.
Raw files	S3. Each PDF is written once and not edited in place.
Ingest trigger	S3 event to SQS. OCR of a long scan stays off Lambda.
Worker	Docker on EC2, IAM role, no keys on the image.
Lake	Parquet, partitioned by drive family.
Table build	Polars writes the code map and the tracing page on each fault row.
Coverage checks	DuckDB over the same Parquet. Which fault codes have no procedure.
Exact codes	BM25. Fault 2310 hits the 2310 row.
Symptom match	Hugging Face embeddings.
Question parse	Named-entity recognition for drive family, fault code, part, and procedure.
Draft	Retrieval-augmented generation from the retrieved rows only.
Cite or refuse	MNLI cross-encoder. A contradiction forces Low.

NLP techniques

Technique	Job
Named-entity recognition	Marks the drive family, fault code, part, and procedure, and filters the index before ranking.
BM25	Matches exact codes and part numbers. Fault 2310 has to hit the 2310 row.
Embeddings	Matches a symptom to the fault text when the strings differ, such as “trips on acceleration” and overcurrent.
Retrieval-augmented generation	Writes technician steps from the retrieved rows only.

Technique	Job
Natural language inference	An MNLI cross-encoder checks that the draft follows the passages. A contradiction forces Low.

OCR sits in front of this, on the worker. It turns a scanned page or a panel photo into text. The tracing page for a fault is a column on the chunk row, written by Polars. It is not a separate graph database.

Big-data placement

Concept	Where it sits
Object storage	Each manual or photo is written once to an S3 raw zone. The original file is not edited in place.
Columnar storage	Parsed pages are Parquet, partitioned by drive family. An ACS880-01 question skips the other frames.
Event-driven ingest	An S3 event places the object on SQS. A Docker worker on EC2 parses it, under an IAM role, with no keys on the image.
Single-node analytics	Polars builds the code map on the worker. DuckDB answers coverage questions over the same Parquet.

The ACS880 English manuals are tens of megabytes, not hundreds of gigabytes. They fit one worker. The lake layout is what keeps a later frame's PDF from being mixed into another frame's answer.

Files to download

English only. On each library page, choose Latest revision. Ignore the 3,138-hit ACS880 search. Those hits repeat languages, old revisions, catalogs, and declarations.

Save these 11 PDFs:

File	Document ID
Primary firmware manual, AINLX	3AUA0000085967
Primary firmware manual, YINLX	3AXD50001000998
ACS880-01 hardware	3AUA0000078093
ACS880-07 hardware, 45 to 710 kW	3AUA0000105718
ACS880-04 hardware	3AUA0000138495
ACS880-04XT hardware	3AXD50000025169
ACS880-14 and ACS880-34 hardware	3AXD50000022021
ACS880-104 hardware	3AUA0000104271
ACS880-104LC hardware	3AXD50000045610
ACS880-107 hardware	3AUA0000102519
ACS880-107LC hardware	3AXD50000196111

Each file is at https://library.abb.com/d/ followed by the document ID.

Open each link list below and download the English hardware PDF. Skip it if that file is already one of the 11.

Frame	Document ID
ACS880-04, 200 to 710 kW	9AKK105713A4819

Frame	Document ID
ACS880-07, 560 to 2800 kW	9AKK105713A6663
ACS880-07CLC	9AKK107046A0239
ACS880-07LC	9AKK107680A9275
ACS880-11	9AKK106930A9565
ACS880-17, 45 to 400 kW	9AKK106930A3466
ACS880-17, 160 to 3200 kW	9AKK106354A1499
ACS880-17LC	9AKK107492A4721
ACS880-31	9AKK106930A9564
ACS880-34, 132 to 400 kW	9AKK107045A8025
ACS880-37, 45 to 400 kW	9AKK106930A3467
ACS880-37, 160 to 3200 kW	9AKK106354A1500
ACS880-37LC	9AKK107492A4722

9AKK105408A7004 and 9AKK105408A8149 are the link lists for the ACS880-01 and ACS880-07 manuals already in the table of 11. Crane, winch, fieldbus, and FSO safety manuals are separate books. Until that PDF is in the lake, a question about that option is Low.

Fault codes in the primary control program

Codes are hexadecimal. This list is the fault table in firmware manual 3AUA0000085967, as far as the extracted text goes. The extract stops at 7301. Warnings use a separate Axxx series and are not listed here. Supply-unit code 2E01 is not an inverter fault.

Code	Fault	Code	Fault
2281	Calibration	5080	Fan
2310	Overcurrent	5081	Auxiliary fan broken
2330	Earth leakage	5090	STO hardware failure
2340	Short circuit	5091	Safe torque off
2381	IGBT overload	5092	PU logic error
2391	BU current difference	5093	Rating ID mismatch
2392	BU earth leakage	5094	Measurement circuit temperature
3130	Input phase loss	5681	PU communication
3180	Charge relay lost	5682	Power unit lost
3181	Wiring or earth fault	5690	PU communication internal
3210	DC link overvoltage	5691	Measurement circuit ADC
3220	DC link undervoltage	5692	PU board powerfail
3280	Standby timeout	5693	Measurement circuit DFF
3291	BU DC link difference	5694	PU communication configuration

Code	Fault	Code	Fault
3381	Output phase loss	5695	Reduced run
3385	Autophasing	5696	PU state feedback
4000	Motor cable overload	5697	Charging feedback
4210	IGBT overtemperature	5698	Unknown power unit fault
4290	Cooling	6000	Internal SW error
42F1	IGBT temperature	6181	FPGA version incompatible
4310	Excess temperature	6306	FBA A mapping file
4380	Excess temperature difference	6307	FBA B mapping file
4381	PCB space cooling	6481	Task overload
4981	External temperature 1	6487	Stack overflow
4982	External temperature 2	64A1	Internal file load

Code	Fault	Code	Fault
64A2	Internal record load	6885	Text file overflow
64A3	Application loading	7080	Option module comm loss
64A5	Licensing fault	7081	Panel port communication
64B0	Memory unit detached	7082	Ext I/O comm loss
64B1	Internal SSW fault	7121	Motor stall
64B2	User set fault	7181	Brake resistor
64E1	Kernel overload	7183	Brake resistor excess temperature
6581	Parameter system	7184	Brake resistor wiring
65A1	FBA A parameter conflict	7191	Brake chopper short circuit
65A2	FBA B parameter conflict	7192	Brake chopper IGBT excess temperature
6681	EFB comm loss	71A2	Mechanical brake closing failed
6682	EFB config file	71A3	Mechanical brake opening failed
6683	EFB invalid parameterization	71A5	Mechanical brake opening not allowed
6684	EFB load fault	71B1	Motor fan
6881	Text data overflow	7301	Motor speed fault; the name is cut off in the extract
6882	Text 32-bit table overflow	7310	Overfrequency trip, named on a parameter page, not in the extracted fault table
6883	Text 64-bit table overflow		

The current PDF of 3AUA0000085967 is the complete fault chapter. Index that file. Do not treat this extracted list as a substitute for the manual. Tag AINLX chunks and YINLX chunks separately so a step from one firmware package is not shown for the other.